When Bihar Went Dry

Using difference-in-differences to figure out if the 2016 alcohol ban actually worked

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The policy, and the question it hides

On 5 April 2016, Bihar stopped being a place where you could legally buy alcohol. The manufacture, sale, transport, storage and consumption of alcohol, all became illegal at once, across a state of more than a hundred million people. There were no slow phase-in and no patchwork of dry districts to ease into it. Chief Minister Nitish Kumar had announced the plan only the previous November, the legislature passed it unanimously, and within months all the shops that served alcohol were gone.

For anyone who cares about cause and effect, a policy like this is almost a gift. Most of the time the interventions we want to study show up gradually, or apply to some people and not others. Bihar's ban wasn’t one of that. It hit everyone in the state on the same morning.

And the precise question is not the one the newspapers asked “Did things change in Bihar after 2016 the answer is obviously yes. The harder and more useful question is how much of that change the ban actually caused. Drinking, crime, road accidents, household spending all of these move around for a dozen reasons that have nothing to do with alcohol prohibition. Finding how the ban of alcohol has affected these things is the much harder question.

Why it isn’t as simple as comparing before and after the ban

The instinct is to line up Bihar before the ban against Bihar after the ban and read off the difference. The simple logic one would come up with would be to compare male drinking for different years and if it decreases, the prohibition worked. The problem is that we don’t know about the version of Bihar that never banned alcohol.

Over the same timeline, a lot was happening to the whole country at once. Incomes were rising, survey methods were changing, national attitudes to drinking were shifting, road-safety rules were tightening. A plain before-after comparison credits every one of those background trends to the ban. If drinking was already drifting down across India, some of Bihar's fall would have happened anyway, and we would be handing prohibition a medal it didn't earn. This is the main problem of causal inference.

The concept of difference-in-differences

Difference-in-differences (DiD) gets around this by borrowing a counterfactual from somewhere else. The move is simple to state. Find a comparison group that was not subject to the ban but was plausibly on the same trajectory as Bihar. Watch what happens to that group over the same window. Whatever drift it shows is your estimate of the drift Bihar would have had anyway. Subtract it, and what's left is the part that only Bihar experienced because of the ban.

There are two differences in the name, and it pays to keep them straight. The first is the treated group's own change, after minus before. The second is the control group's change over the identical period. The causal estimate is the first difference minus the second:

$$DiD = (Bihar_{02} - Bihar_{01}) - (Control_{02} - Control_{01})$$

The same thing can be written as a regression, which is how I actually estimated it. Code Bihar as 1 and the control as 0, code the post-ban period as 1 and the pre-period as 0, and run:

$$Y = \beta_0 + \beta_1 \cdot Bihar + \beta_2 \cdot Post + \beta_3 \cdot (Bihar \times Post) + \varepsilon$$

Here β_1 soaks up the permanent gap between the two states, β_2 soaks up whatever happened to both of them over time, and the interaction term β_3 is the bit we care about which is the extra movement in Bihar, on top of the common trend, in the post-ban period. That coefficient is the difference-in-differences estimate of the causal effect.

All of this rests on one assumption. We assume parallel trends that would've happened in absence of the ban. We can't prove it given that it's a statement about a world that didn't happen but we can make it more or less believable by choosing the control carefully. Which is the next decision and one that matters the most.

Picking the control: why Jharkhand

A DiD study is only as good as its control group, so this is not a random choice. Comparing Bihar to "all of India" would not make sense: the national average folds in rich coastal states and far-off regions with nothing in common with Bihar. I picked a neighbour that was as close to a twin among other Indian States.

Jharkhand fits well, because until recently it literally was Bihar. They were one state until November 2000, when Jharkhand was carved out of the southern districts. They share a border, overlapping language and culture, and similar levels of income and development. If any state was going to drift the way Bihar drifted in the absence of a ban, it has to be Jharkhand. That shared past is the strongest argument I could make for parallel trends.

Jharkhand also kept alcohol legal throughout, which is the other thing a control needs to do: stay untreated. So we'll consider Bihar as treatment and Jharkhand as control, with the break at April 2016 splitting "before" from "after."

The leak in the design, and the fix

Jharkhand isn't a sealed laboratory sitting safely next door. The moment Bihar went dry, the obvious workaround for a Bihari who wanted a drink was to drive across the border and buy it in Jharkhand. We don't have to guess whether this happened, the money says it all. Jharkhand's excise revenue climbed from about ₹912 crore in 2015–16 to roughly ₹1,600 crore by 2017–18, a jump that is hard to explain without the cross-border custom.

If Bihar's ban changes behaviour in Jharkhand, Jharkhand is no longer a clean control for the Bihar treatment. What makes this subtle is that the bias runs in different directions depending on which outcome you look at, because the three outcomes are measured differently.

Take drinking first. A Bihari man who drives to Jharkhand to drink is still surveyed back home in Bihar, and still counts as a Bihar drinker. So that leak mostly makes Bihar's measured fall look smaller than the true fall, which works against finding an effect. That is the safe direction to be wrong in. Crime and road accidents are the opposite. Those get counted where they happen. If an intoxicated Bihari crashes or brawls on the Jharkhand side of the line, the incident lands in the control group's totals, pushing them up, which makes the DiD look more negative than it should and overstates the ban's benefit.

The standard remedy is a “donut” design that is to exclude the districts that share the border, where the leakage concentrates, and compare only the interiors of each state, where a thirsty resident is too far from the line for a quick run across the borders. So I’ve dropped eight Bihar districts that touch Jharkhand (Rohtas, Aurangabad, Gaya, Nawada, Jamui, Banka, Bhagalpur and Katihar), along with the Jharkhand districts on the other side of that seam: Garhwa, Palamu, Latehar, Chatra, Hazaribagh, Koderma, Giridih and Dumka. What's left is interior Bihar against interior Jharkhand.

The most careful crime study of the ban finds that the drop in violent crime is concentrated precisely in the interior districts and fades toward the border. That is exactly the fingerprint you'd expect if the border districts are diluted by leakage. The contamination we're worried about leaves a visible mark, and trimming the border is the right response to it.

Did people actually stop drinking?

Everything else downstream depends on this first. A ban on paper means nothing if people kept drinking exactly as before through smuggled or home brewed liquor. So, the first model asks the most basic question which is - after April 2016, did fewer men in Bihar drink?

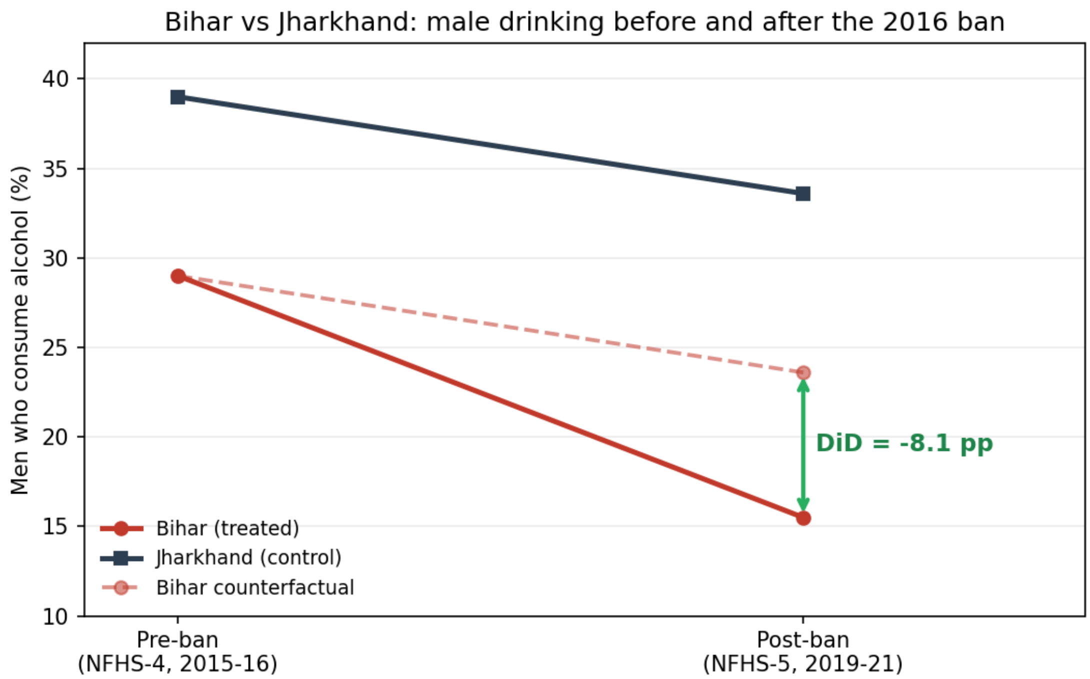
The cleanest measure comes from the National Family Health Survey, which asks a large representative sample of men whether they consume alcohol. NFHS-4 was fielded in 2015–16 — in Bihar, between March and August 2015, comfortably before the ban — and NFHS-5 in 2019–21, well after. That gives a pre ban and post ban scenario for both states. The four numbers are as follows –

Men who consume alcohol (%)	Pre (NFHS-4)	Post (NFHS-5)	Change
Bihar (treatment)	29.0	15.5	**−13.5 pp**
Jharkhand (control)	39.0	33.6	**−5.4 pp**

Notice that drinking falls in both states. Some of that is because of the ban and some of it isn’t. The survey widened its age definition between rounds, which mechanically lowers the later numbers. But this is the whole point of the method. The common force that pulled both states down is captured by Jharkhand's 5.4-point slide and then subtracted out. What survives is Bihar's extra fall:

$$\textit{DiD} = (-13.5) - (-5.4) = -8.1 \textit{ percentage points}$$

So the ban is associated with roughly an eight-point larger drop in male drinking in Bihar than the common trend can explain. Running the four cells through the regression returns $\beta_3 = -8.10$. The figure below is the canonical way to see it: the gap between where Bihar actually landed and where Jharkhand's trend says it should have landed is the effect.



Is the gap just noise?

Four cell means give a point estimate but no standard error, so on their own they can't tell us whether 8 points is real or a fluke. To show how the inference actually goes, I rebuilt individual-level records from the published prevalences and the (approximate) NFHS male sample sizes (a few thousand simulated respondents per state-round, each a one or a zero), then ran the same DiD as a linear-probability regression. The interaction came back at about −8.4 points with a standard error near 1.2 points and a p-value down around 10^{-12}. A gap that size, with samples that big, is almost impossible.

After 2016, admitting to drinking in Bihar means admitting to a crime, and people understandably clam up. If some Bihari men kept drinking but stopped saying so, the survey will record a fall that is partly behaviour and partly silence. That would make the measured 13.5-point drop an overstatement of the real behavioural change. It doesn't sink the result — the DiD is still picking up something large and Bihar-specific — but it's a reason to read the alcohol number as an upper bound on how much actual drinking fell.

Model 2 — Did less drinking mean less crime?

If the ban really cut drinking, the next interesting question to ask is whether the social costs of drinking fell with it. There's a clean theoretical reason to expect alcohol to matter more for some crimes than others. Alcohol fuels aggression and clouds judgement, so it should bear on violent crime far more than on premeditated property crime. That asymmetry is itself a test. If prohibition cut violent crime but left property crime alone, the drinking channel is the likely story. If it moved everything equally, you'd suspect something like a general change in policing.

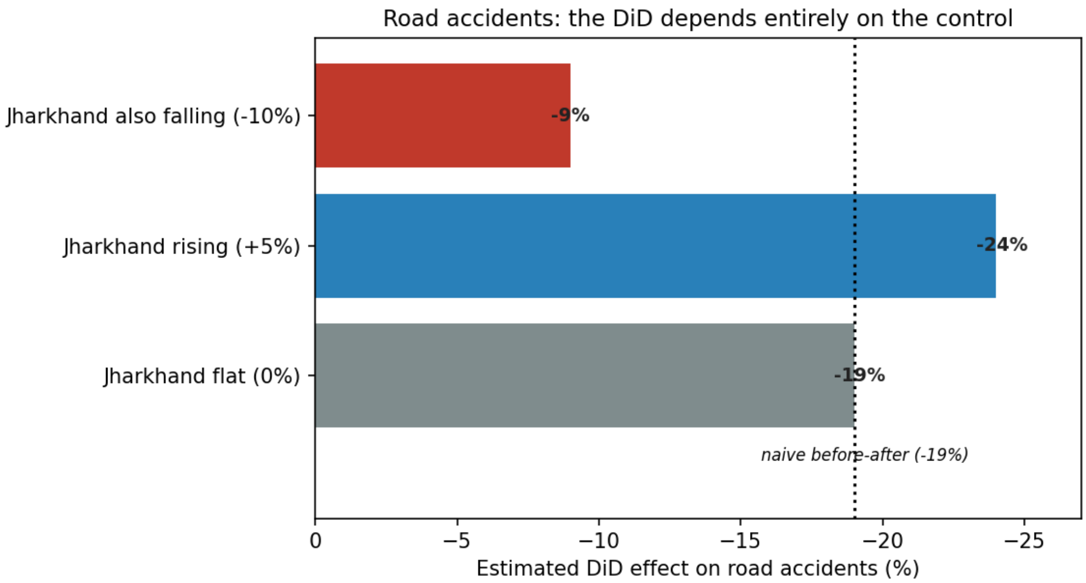
The most careful study of this, done by Chaudhuri and co-authors in Economic Development and Cultural Change, runs exactly this kind of DiD on a district-by-month panel of reported crime stretching from January 2013 to February 2019. Their headline is a reduction of about 0.22 of a standard deviation in reported violent crime, with no meaningful effect on non-violent crime, and the effect holds up across the three years following the ban rather than fading after the initial shock. I would like to use their estimate here rather than using my own because doing it properly needs their full district-month panel, reproducing the arithmetic of the 2×2 is easy, but the credibility lives in that panel.

What I find genuinely persuasive is where the effect shows up. It's larger in interior districts and in districts that drank more heavily to begin with, and larger where fewer people faced religious restrictions on alcohol in the first place. Every one of those is a place where the ban should have bitten hardest on actual drinking. If the crime drop were really just stricter policing dressed up, it would be smeared evenly across the state. Instead it clusters exactly where drinking fell the most which points to the drinking channel as the mechanism and approves the donut logic from earlier. The interior is where the signal is strong and the border is where it leaks away.

Model 3 — Road accidents, and a cautionary tale

The third outcome is the one the government was keenest to advertise. In the months after the ban, officials reported that the road accidents in Bihar were down by about 19 percent and accident deaths were down by about 31 percent against the same months a year earlier. Drunk driving is a real killer, so a fall is entirely plausible. The problem is what kind of number that is.

A 19-percent fall measured as Bihar after versus Bihar before is a pure first difference, the before after comparison we already agreed isn't enough. To turn it into a DiD one would have to subtract whatever was happening to road safety in a comparable place(Jharkhand in this case) over the same window, and the answer depends entirely on what that comparison did.



If road safety in Jharkhand was improving anyway over those years — better roads, helmet enforcement, the same national push that was bringing accidents down in a lot of places, then a good chunk of Bihar's 19 percent would have happened without any ban, and the true causal effect could be as small as nine points. If accidents were drifting up in the control, the ban looks even better. And the spillover problem we met earlier comes again. Drunk Bihari drivers who now crash on the Jharkhand side push the control's accident count up, which mechanically drags the DiD downward and flatters the ban. Without a clean interior district accident panel for both states, which isn't published at the granularity, this outcome stays the most weakly identified of the three.

Reading the three together

The three models tell a coherent story rather than three disconnected ones. The alcohol result is the first stage. The ban genuinely cut how many men drink, by something like eight points beyond the common trend, and that estimate sits on the firmest data of the three. Crime is the downstream effect, violent crime fell, non-violent crime didn't, and the fall is concentrated exactly where drinking dropped the most which is what one would predict if it's the ban of alcohol doing the work. Road accidents are the plausible but shaky third leg, pointing the same way but resting on a before-after number that a proper control could shrink by half.

Outcome	DiD estimate	How much to trust it
Male alcohol use	−8.1 pp	Strongest, real NFHS 2×2, watch for under-reporting under a ban.
Violent crime	−0.22 SD	Solid, peer-reviewed, concentrated in interior, heavy-drinking districts.
Road accidents	−9% to −24%	Weakest, a before after headline, the real DiD hinges on the control.

What could still be wrong

Parallel trends is an assumption, not a fact, and with only one pre period and one post period in the alcohol data, one cannot actually plot the two states moving together beforehand. The shared history makes it believable, but believable isn't exactly proven.

Then there's measurement, which cuts at every outcome. Self-reported drinking under a prohibition regime is exactly the kind of thing people shade, so part of the measured fall is silence. Reported crime is filtered through police behaviour, and a prohibition that floods the courts with liquor cases and redeploys officers can change how ordinary crime gets recorded. The NFHS age definition changes between rounds shifts the levels even though the DiD cancels out anything common to both states. And spillover, even after the donut model, never fully disappears. Trimming the border reduces the leak but doesn't stop it.

The cleanest version of this study would add a second comparison state or build a synthetic control out of several, run a proper event study to eyeball the pre-trends and use the underlying district month microdata for standard errors. Based on the numbers and the data that I could work with, the alcohol model is the one I can stand fully behind on its own data, crime borrows its credibility from the published panel study and road accidents is deliberately presented as the example of what happens when the counterfactual is missing.

So what did we actually learn

The first thing we learnt is about Bihar. The 2016 ban did more than exist on paper. It cut self reported male drinking by a huge margin, violent crime fell in step and in the right places, and accidents came down too, though by how much is unclear. That's a real policy effect.

The second is about the method. Difference-in-differences doesn't give certainty, what it does is force the counterfactual into the open and make you argue for it. The interesting work was not the subtraction, that is a line of arithmetic. It was choosing Jharkhand and then immediately interrogating that choice. Noticing the cross-border leak, signing which way it biased each outcome, and trimming the border to deal with it. A before after number hides all of that and hands you a tidy headline. A difference-in-differences makes you earn the headline, and tells you which ones you haven't earned yet. On this evidence, the drinking and crime results are earned. The road accident one still owes us a control.